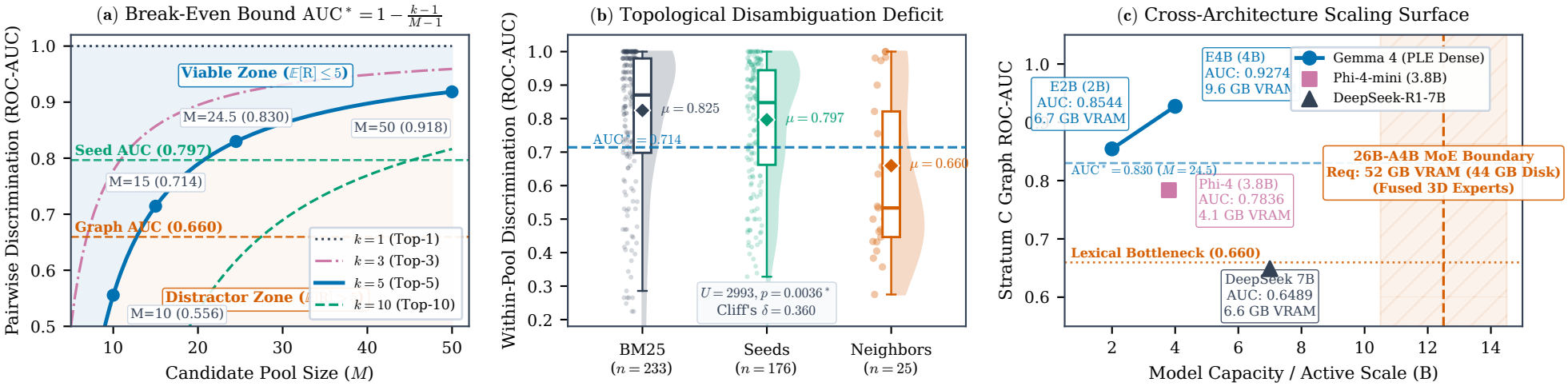

(a) Break-Even Bound $\mathrm{AUC}^* = 1 - \frac{k-1}{M-1}$
Viable Zone ($\mathbb{E}[\mathrm{R}] \leq 5$)
M=24.5 (0.830)
M=50 (0.918)
Seed AUC (0.797)
M=15 (0.714)
Graph AUC (0.660)
Distractor Zone
M=10 (0.556)
$k=1$ (Top-1)
$k=3$ (Top-3)
$k=5$ (Top-5)
$k=10$ (Top-10)
Pairwise Discrimination (ROC-AUC)
Candidate Pool Size (M)
(b) Topological Disambiguation Deficit
$\mu = 0.825$
$\mu = 0.797$
$\mu = 0.660$
AUC = 0.714
$U = 2993, p = 0.0036^*$
Cliff's $\delta = 0.360$
BM25 ($n = 233$)
Seeds ($n = 176$)
Neighbors ($n = 25$)
Within-Pool Discrimination (ROC-AUC)
(c) Cross-Architecture Scaling Surface
E2B (2B)
AUC: 0.8544
6.7 GB VRAM
E4B (4B)
AUC: 0.9274
9.6 GB VRAM
Gemma 4 (PLE Dense)
Phi-4-mini (3.8B)
DeepSeek-R1-7B
$\mathrm{AUC}^* = 0.830$ ($M = 24.5$)
Phi-4 (3.8B)
AUC: 0.7836
4.1 GB VRAM
26B-A4B MoE Boundary
Req: 52 GB VRAM (44 GB Disk)
(Fused 3D Experts)
Lexical Bottleneck (0.660)
DeepSeek 7B
AUC: 0.6489
6.6 GB VRAM
Stratum C Graph ROC-AUC
Model Capacity / Active Scale (B)